

PoS P-Quiz (Philosophy of Science)

- **Quiz Type:** Practice Quiz
 - **Total Points:** 27
 - **Questions:** 27
 - **Due Date:** May 10, 2026
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Question 1 (1 pt)

Which of the following is an example of an 'internal virtue' of a theory? ☒ The theory is compatible with other well-established scientific beliefs. ☐ The theory accurately predicts the outcome of a useful new experiment. ☐ The theory leads to the development of a useful new technology. ☐ The theory successfully explains a puzzling natural phenomena.

Question 2 (1 pt)

Which of the following best describes the relationship between the concepts of 'theory', 'hypothesis', and 'law'? ☒ They describe different, non-exclusive features of a scientific claim: its subject matter, its justification status, and its generality. ☐ They are mutually exclusive; a claim can be a theory, a hypothesis, or a law, but more than one at a time. ☐ A claim progresses from a hypothesis to a theory and finally graduates to the status of a law once it is proven. ☐ A law is a mathematical theory, while a hypothesis is a non-mathematical theory.

Question 3 (1 pt)

What is the central problem of 'underdetermination' in philosophy of science? ☐ Scientists often lack the imagination to come up with enough creative theories to explain all the available data. ☐ Experiments are always too complex and expensive to provide conclusive proof for any single theory. ☒ The available observational evidence is not sufficient to uniquely determine which of several competing, empirically equivalent theories is true. ☐ A theory cannot be considered scientific until it has been determined to be true beyond all possible doubt.

Question 4 (1 pt)

Select the statement that best describes the concept of 'internal virtues' of a theory. ☐ Internal virtues are features that indicate that the theory is compatible with experimental results, i.e., that the theory is accountable to observations. ☐ Internal virtues are non-epistemic features of a theory, including social, aesthetic, and practical values. ☐ Internal virtues are hierarchical features of a theory, e.g., simplicity and generality are at the top, followed by testability, explanatory cooperation, and entrenchment at the bottom. ☒ Internal virtues are features of a theory that are true-conducive and that make it better at producing true, reliable, and justified beliefs.

Question 5 (1 pt)

In the context of empirical assessment, what is the typical pragmatic distinction made between the activities of 'confirmation' and 'explanation'? ☒ Explanation typically covers past observations of events whose causes were puzzling, while testing is reserved for future observations of predictions whose outcomes are yet unknown. ☐ Confirmation reports contact with the world, whereas explanation is an

internal feature of a theory's coherence. ☐ Confirmation focuses on predicting the consequence of the theory, while explanation focuses on the causal structure of the universe. ☐ Explanation provides inductive support for a theory's truth, while confirmation provides only psychological satisfaction.

Question 6 (1 pt)

Select the statement that best describes the concept of 'hypothesis'. ☐ The concept 'hypothesis' is used to identify the longest side of a right-angled triangle, located directly opposite the 90-degree right angle. ☐ A hypothesis is a vague thought. ☒ The concept 'hypothesis' is used to identify a particular theory (or theories) that is in need of testing. ☐ A hypothesis is a claim about the external world.

Question 7 (1 pt)

"All copper conducts electricity" is an example of what? ☒ A law ☐ An accidental association ☐ A hypothesis ☐ The wonders of metallurgy

Question 8 (1 pt)

Non-epistemic virtues of a scientific theory are ... ☐ philosophical dilemmas. ☒ qualities that contribute to a theory's usefulness, social relevance, or practical application but are not directly related to its truth or empirical adequacy. ☐ components in theories of knowledge. ☐ features that make a scientific theory better at producing true, reliable, and justified beliefs.

Question 9 (1 pt)

What is one of the problems of the covering-law model of explanation? ☐ It includes the problem of not covering every law. ☒ It includes the problem of distinguishing laws from accidental generalizations. ☐ It includes the problem of not covering every condition. ☐ It includes the problem that the laws differ from country to country.

Question 10 (1 pt)

What makes an explanation a good explanation? ☒ The features (i.e., the truth-conducive virtues of the explanation) cannot be evaluated without reference to some theories, and it must be clear that the law and the conditions are describing relevant factors and not just circumstantial properties. ☐ When all scientists agree with it. ☐ The features (i.e., the truth-conducive virtues of the explanation) can be evaluated without reference to some theories, and it must be clear that the law and the conditions are describing circumstantial properties. ☐ The features must contain non-epistemic virtues, which are true-conducive.

Question 11 (1 pt)

Which of the following statements describes the D-N model the best? ☐ The D-N model is an inductive argument that involves specific conditions and at least one law of nature. ☐ The D-N model is a deductive argument that involves specific conditions but no laws of nature. ☒ The D-N model is a deductive argument that involves specific conditions and at least one law of nature. ☐ The D-N model is one of Pakistan's most trusted brands.

Question 12 (1 pt)

Which of the following is not a problem of the covering-law model of explanation? ☐ Problem of Symmetry ☒ Problem of Adequacy ☐ Problem of Relevance ☐ Problem of Conjunction

Question 13 (1 pt)

Which of the following statements best describes the concept of 'hypothetic-deductive (H-D) confirmation'? ☒ This is a very general, formal model of confirmation in which a hypothesis is tested by its deductive consequences. ☐ This is a very general, formal model of confirmation in which a hypothesis is tested by its inductive consequences. ☐ This is a very general, formal model of confirmation in which a hypothesis is tested by its bioconditional consequences. ☐ This is a very general, formal model of confirmation in which a natural law is tested by its deductive consequences.

Question 14 (1 pt)

Is the following statement correct? *"There is an asymmetry in the basic model of confirmation. Confirmation comes in degrees and is vulnerable to confusion as to how much is enough. Falsification, on the other hand, shows no such sloppiness, no degrees, no need for discretion. A single failed predication will suffice to falsify a theory and force its rejection."* ☒ Correct ☐ Incorrect

Question 15 (1 pt)

The revised hypothetico-deductive model contains three components. They are: ☐ The laws, the hypothesis, and the predicted effect. ☐ The theories, the laws, and the

predicted effect. ☒ The hypothesis, the conditions of testing, and the predicted effect. ☐ The hypothesis, the conditions of testing, and the experiment.

Question 16 (1 pt)

Which statement best describes the concept of an 'auxiliary theory'? ☐ An auxiliary theory is a supporting theory that helps make a scientific experiment successful. ☒ An auxiliary theory is the background knowledge required to draw testable consequences from a hypothesis. ☐ An auxiliary theory is a special form of hypothesis. ☐ An auxiliary theory refers to the concept of 'prediction.'

Question 17 (1 pt)

Instrumentalism is a branch of ☐ realism ☒ antirealism ☐ ethics ☐ aesthetics

Question 18 (1 pt)

An instrumentalist claims that theories ☐ are useless. ☐ always true. ☒ are intellectual tools for the purpose of prediction of future phenomena and organisation of observation. ☐ cannot be evaluated in terms of applicability.

Question 19 (1 pt)

Is the following statement correct? *"An instrumentalist thinks that unobservable entities, such as quarks, should be taken literally."* ☐ Yes ☒ No ☐ Maybe ☐ None of the above.

Question 20 (1 pt)

Empiricists think that ☐ that unobservables can only be known by reason. ☐ that we can have justified knowledge only about unobservables. ☒ that we can have justified knowledge only about observables. ☐ that unobservables do not exist at all.

Question 21 (1 pt)

The position of the empiricist demands ☐ atheism towards unobservables. ☒ agnosticism towards unobservables. (*Note: empiricism holds agnosticism regarding unobservable entities*). ☐ strong belief in unobservables. ☐ sympathy towards unobservables.

Question 22 (1 pt)

Realism assumes that ☐ we cannot say anything about unobservables. ☒ our best scientific theories (aim to) give true or approximately true descriptions of observables and unobservable aspects of the world. ☐ we should endorsing talking about unobservables like electrons and atoms as true. ☐ the world is an illusion.

Question 23 (1 pt)

The link between empirical adequacy and truth of a theory is known as ... ☐ deductive argument. ☒ inference to the best explanation. ☐ inductive argument. ☐ hermeneutical circle.

Question 24 (1 pt)

Inference to the best explanation can be defined as: ☒ The causal explanation that is most fruitful, that is, explains the most phenomena with a minimum of theoretical extravagance, and that fits into the larger theoretical network, is more likely to be true than a theory that does none of this or does only some of this. ☐ As the attitude of doubting knowledge claims, questioning whether certainty is possible, or demanding evidence before accepting beliefs ☐ As a "top-down" logical process where a specific, guaranteed conclusion is drawn from general premises or rules. ☐ As a logical process where specific observations, experiences, or data patterns are used to form broader, general conclusions or hypotheses.

Question 25 (1 pt)

Which of the following is the most observable? ☒ Books ☐ Fleas ☐ Cells ☐ DNA molecules atoms

Question 26 (1 pt)

Which of the following is not a component of an 'accountable observation'? An accountable observation must be: ☐ informative ☐ assertive ☒ obscure ☐ propositional

Question 27 (1 pt)

Entrenchment as an internal virtue of a theory refers to: ☒ A theory's degree of compatibility and coherence with already well-established theories in science. ☐ A theory being immune to testing. ☐ A theory that has only mathematical equations. ☐ A theory verified purely by intuition.